

Architecture Justification

Project: Digital Art Commission & Watermarking Portal

Architecture Selection

We chose **Layered Architecture** for the Digital Art Commission & Watermarking Portal.

1. Reasons for Choosing Layered Architecture

Separation of concerns: The system can separate the user interfaces, commission management logic, and supporting services such as payment, watermarking, file storage, and database access. This makes the system easier to understand and maintain.

Modularity and maintainability: Services such as the Payment Service and Watermark Service can be changed or improved without major changes to the client or artist interface. This suits the portal because it has separate responsibilities for commissions, payments, watermarking, and file delivery.

2. Security Advantage

Sensitive operations such as milestone payments and final high-resolution file downloads are handled by dedicated services rather than directly through the user interface. This supports better access control and helps protect artwork and payment-related information. Expiring signed URLs can also be used for final downloads.

3. Performance Advantage

Separating services such as watermark processing and file delivery allows these operations to be handled independently. This can improve response time when clients review WIP drafts or download final high-resolution artwork while keeping the main commission workflow responsive.

Conclusion: Layered Architecture is suitable because it keeps the portal organized, maintainable, and easier to secure while supporting its main commission workflow.